

Evidentiary reasoning with the Analytical Hierarchy Process: Use of graph entropy to rank decision elements

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GitHub: hamlett-neil-ur/ahp-graph-entropy

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Abstract

Decision makers often require insight into the relative strength of evidence influencing their choices. The Analytic Hierarchy Process (AHP) contains innate mechanisms leading to the rigorous quantification of evidentiary elements' relative evidentiary strength. Pairwise-comparison analysis produces lists of edges for a complete, directed, acyclic graph, often called a tournament graph. The resulting *Intensities of Importance* provide tournament-graph edge weights, leading to a novel Laplacian matrix. Relative-entropy calculations based on Laplacians for induced subgraph quantify the relative evidentiary strength associated with the decisional evidence omitted from them. A previously published case study related to a fuel-cell design demonstrates the approach. The graph-entropy method provides clearer strength-of-contribution indications than sensitivity analysis from the case study.

Introduction.

Decision-making nearly always occurs under conditions of incomplete or imperfect information. No less than Nobel-prize economist Frederick von Hayek recognized this:

...I prefer true but imperfect knowledge, even if it leaves much undetermined and unpredictable, to a pretence of exact knowledge that is likely to be false (emphasis added) (von Hayek (1974), cited in Kay & King (2020)).

The treatment of uncertainty in quantitative decision making represents a formidable research challenge confronting multiple facets. We conventionally decompose

uncertainty into epistemic and aleatory (or stochastic) components (Salicone & Prioli (2018)). Sound evidentiary reasoning must account for the consequences of both uncertainty components.

The present study focuses primarily on the epistemic. Hamlett (2019, 2025) argues for the primacy of resolving epistemic uncertainty in business decision making. Exemplified by Zellweger & Zenger (2023); Rindova & Martins (2024); Townsend, et al (2025); Ehrig, et al (2026), this perspective holds sway in the entrepreneurial and strategic decision-making communities. Jackson (2019) corroborates from a philosophy-of-science point of view.

Motivation: How strong is our evidence?

Ehrig, et al (2026) argue that entrepreneurial decision making in particular inescapably requires "...logic and reasoning with the aid of very small samples of data" (emphasis added). Circumstances demanding this sort of reasoning arise in numerous contexts beyond new-business formation (Martin & Olsberg (2007); Eisenmann (2013); Swanson (2024)). Fundamentally, entrepreneurial ventures amount to epistemic explorations (Camuffo, et al (2024)). Howard & Abbas (2016) devoted considerable attention to the adverse effects of imperfect information — an endemic condition within entrepreneurship — when trying to select optimum scenarios.

Nontrivial challenges inevitably arise even when considerable amounts of statistically coherent, phenomenologically illuminating data exist. Cover & Thomas (2006) derive an information-theoretic data-processing inequality: "...no processing of Y ... can increase the information that Y contains about X ." Corroborating from a statistical-learning perspective, Hastie, Tibshi-

rani, Friedman (2009) define *irreducible variance*. Kay & King (2020) warn of pervasive *nonstationarity* in real-world problems: Statistical-learning training data based on past experience might not represent future conditions very well. More generally, Townsend, et al (2025) articulate the case for “...computationally irreducible barriers to the predictive capabilities and task performance advantages of AI systems...” for many tasks.

Extending Dempster’s (1967) seminal concept, Shafer (1976) offered a rigorous approach to quantifying epistemic uncertainty in evidentiary reasoning. More-recently, Shenoy (2023) applied Shafer’s reasoning to quantify evidentiary incompleteness in Bayesian networks, a causal-inference methodology commonly employed within the entrepreneurial decision-making community (Camuffo, et al (2024); Ehrig, et al (2026)).

Hué, et al (2026) use a stochastic-detection area-under-the-curve (AuC) approach (cf. Fawcett (2026)) to analyze evidentiary elements’ relative strengths. Resembling stepwise regression (Olive (2017)), their approach also exhibits conceptual parallels to Shafer (1976). Cox (2026a,b) offers an emerging approach to uncertainty characterization based on sheaf theory (Rosiak (2022)).

Why the Analytical Hierarchy Process (AHP)?

AHP provides an axiomatic framework (Saaty (1986)) for “...logic and reasoning with the aid of very small samples of data” (Ehrig, et al (2026)). It is a multi-criterion decision methodology (MCDM) enjoying widespread applicability to group decision making (Saaty (1994)), risk analysis (e.g., Saaty (1987); Vaidya & Sushil (2006); Linkov, et al (2006)), and other problems. AHP is often applied to situations where substantial amounts of statistically coherent measurements are not immediately available to support more-empirical approaches.

The AHP methodology begins with identifying a plausible evidentiary span: The range of measurable factors — elements in the vernacular of Saaty (1986, 1994) — believed to differentiate the relative attractiveness within the range of scenarios under consideration. A hierarchy model represents this as a tree graph. The responsible individual decision maker or panel of experts then performs *comparative judgements*: “...the numerical representation of a relationship between two elements that share a common parent” (emphasis added) in the tree graph (Saaty (1994)). A case study in a subsequent section demonstrates AHP’s use to rank plausible scenarios based on such information.

Not strictly a stand-alone method, AHP can be fused with other approaches. Bayesian techniques allow for subsequent revision of the comparative judgements as

stronger empirical evidence emerges (Merrick, van Dorp, Singh (2005); Li, H., et al (2025)). Li, J. et al (2025) combined back-propagation neural networks with the AHP to design shields for neutron generators. Xiao, et al (2025) explored simulation-optimization approaches to circumvent altogether the elicitation of comparative judgements from experts. Lukinskiy, Lukinskiy, Bazhina (2025); Sáenz-Royo & Chiclana (2026) studied methods to improve the robustness of expert inputs into comparative judgements.

Evidentiary-strength analysis using the AHP.

That the comparative judgements for an AHP hierarchy model define a complete, directed, acyclic graph (DAG) provides the seminal insight underlying this investigation. Complete DAGs belong to a special class of graph called tournament graph (Moon (2015)). In general, “the entropy of a graph is interpreted as its structural information content and serves as a complexity measure” (emphasis added) (Dehmer & Mowshowitz (2011); Mowshowitz & Dehmer (2012)). Structural information content bears direct relevance to the present concern of evidentiary reasoning. Brown, et al (2020) studied the particular characteristics of tournament graphs.

What questions do entropies of AHP-based tournament graph allow us to answer? Endogenous to the AHP formulation itself, relative entropy quantifies the marginal informational contribution by a subset of elements within the evidentiary span. This addresses the objective pursued by Hué, et al (2026). Graph-entropy measures however do not require extensive empirical data spanning of both positive and negative cases, as with Hué’s approach. Congruent with the AHP methodology explored here, Jiroušek, Kratochvíl, Shenoy (2023) use entropy of belief-function graphical models to decompose their elements’ relative evidentiary strength.

By what mechanism do we infer the differential information between hierarchically-coupled AHP hierarchy models? Relative entropy provides a well-understood approach (Cover & Thomas (2006)). Relative entropy for graphs has been well-researched (Wong & You (1985); Liu, et al (2022); Guanlei, Xiaogang, Xiaotong (2025)). Brown, et al (2020) explored entropy differences between same-order tournament graphs with differing directional structures. In our case, the relative entropy of the graph associated with the overall evidentiary span with respect to a constituent induced subgraph quantifies the information lost by omission of elements from the latter.

What innovation is required to define a methodology for AHP-based evidentiary-strength analysis? First, we must formulate an entropy definition for tournament graph with weighted edges. Brown, et al

(2020) analyzed graphs with unweighted (uniformly unit-weighted) edges. Chen, et al (2015), Kuo (2022), and Liu, et al (2022) formulate entropies for regular (non-directional) graph comprised of weighted edges. The intensities of importance from the AHP comparative judgements provide the weights used here.

Second, relative entropies of hierarchically linked AHP subgraph must be coupled. The *Hierarchy* term in AHP refers to a hierarchic deconstruction of decisions into constituent elements. The AHP method produces a set of decision-element comparative judgements — represented by a tournament graph — for each hierarchy model. (Cover & Thomas, 2006, §2.5) describe chain rules for entropy, relative entropy, and mutual information. Such mechanisms offer a rigorous approach to link hierarchically adjacent tournament graph.

Why is “heavy” graph-entropy machinery needed? We might intuitively expect that individual comparative judgements suffice to inform us about the most-influential elements. The AHP’s machinery however accounts for “mutual coupling” between elements (Saaty (1986)). That comparative judgements define a graph — a network — provides the basis of its power. Our graph-entropy machinery accounts for the aggregate effect of all the evidence informing a decision or risk assessment. Individual importance intensities fail to provide this.

Decision Quality as a check on the completeness of evidentiary reasoning.

Decision Quality (DQ) represents a prominent applied decision-analysis framework that emphasizes evidentiary reasoning (Spetzler, Winter, Meyer (2016)). Philosophically congruent with design-of-experiments practice (Cox & Reid (2000)), DQ provides a *de facto* standard of practice for the Society of Decision Professionals (SDP). SDP is a community of applied decision analysts and scientists. Introducing DQ here — independent of the mechanics of AHP — reinforces our logical completeness.

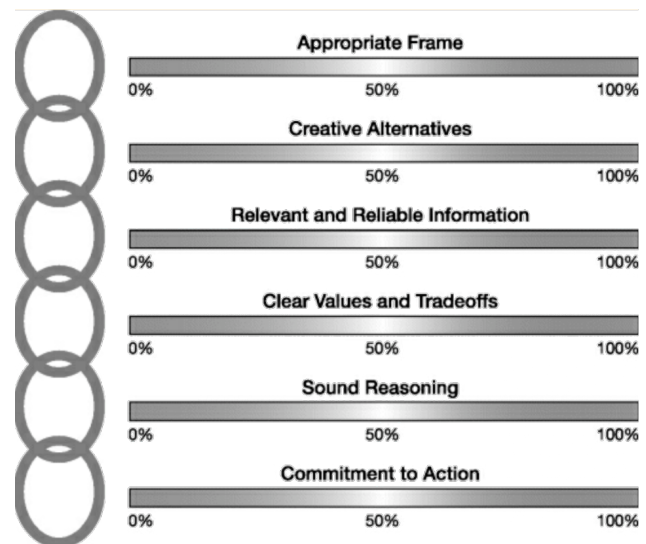
The pattern of reasoning underlying Saaty’s (1986) AHP methodology loosely maps to the DQ framework. Figure 1 depicts the DQ framework. The DQ frameworks contains six *links*:

- Establishing an *Appropriate Frame* specifies the scope of the decision: What specifically is to be decided. Thoughtful framing combines prescriptive with proscriptive judgements to provide analytical clarity. Saaty (1994) describes this as “...the most creative part of decision making that has a significant effect on the outcome...”
- By definition, the act of deciding involves selection of one or more options from a discrete, precisely

defined set of Creative Alternatives.

- *Relevant and Reliable Information* constitute the evidence weighed in selecting the best alternative.
- *Clear Values and Tradeoffs* specify the criteria by which the best alternative is identified.
- *Sound Reasoning* defines the logic employed in identifying the best alternative. Creighton (1909) cautioned that “...it is not possible to lay down definite rules for the guidance of thinking in every case.” The reasoning approach must therefore fit the circumstances resulting from the links above. AHP is itself a distinct reasoning approach. ⚠
- *Commitment to Action* represents the follow-through in implementing or adopting the selected alternative. Absent a commitment to adopt or act upon the best alternative resulting from the links above, a decision arguably has not occurred. The analysis was academic.

Figure 1: Decision-Quality framework. (From Spetzler, Winter, Meyer (2016), Figure 2.1)



Course of investigation.

We begin with a brief illustration of AHP methodology. A *real-world* case study concisely demonstrates its application to obtaining a preference-ranked set of scenarios. The actual ranking lies beyond our specific interest, but is included for completeness. The DAGs resulting from the comparative judgements provide our point of departure into a novel methodology.

We then calculate the entropies for our comparative-judgement DAGs associated with distinct hierarchy

models. This extends Brown, et al (2020) through adoption of a Laplacian structure that includes edge weights. Our AHP-graph Laplacian fuses the structure of a conventional unit-edge-weighted DAG Laplacian with comparative-judgement importance intensities, which provide the edge weights. This incorporates information about degree of influence in the interconnected evidentiary framework.

Finally, we calculate entropies and relative entropies from our AHP-based DAG structure comprised of multiple, hierarchically coupled hierarchy models. We interpret the results for our real-world case study. Approaches analogous to those by Jiroušek, Kratochvíl, Shenoy (2023) guide our thinking. We also explore the relationship between importance intensities and relative entropy. This quantifies relative strengths of contribution by subsets of evidentiary elements.

AHP case study .

Bu, et al (2026) employed AHP in making tradeoff decisions during the design of Proton Exchange Membrane Fuel Cells (PEMFCs), which “...efficiently convert chemical energy from fuels into electricity without causing pollution...”. Bu, et al (2026) contains device-physics technical details that are tangential to our purposes. The AHP mechanics employed by the former primarily interests us.

Bu, et al (2026) particularly suits the purposes of this investigation for three reasons.

1. *It demonstrates a real-world application.* Introductory AHP presentations often employ contrived cases. A real-world case study provides a more-informative and -convincing demonstration.
2. *This case is recent.* Given that the AHP literature spans decades, a reader might succumb to the temptation to conclude that the method is anachronistic. Our case here demonstrates the application of AHP to a contemporary design problem for an economically essential application.
3. *It strikes a balance between tractability and complexity.* Many pedantic demonstrations are borderline-trivial, constructed to illustrate AHP mechanics. The present investigation benefits from a moderately greater degree of complexity. The selected case provides the opportunity to demonstrate the use of relative entropy to couple hierarchically adjacent hierarchy models.

The following offers neither criticism nor endorsement of Bu, et al (2026). Their work merely serves to illustrate AHP mechanics.

We strive in the following to select AHP terminology that remains close to Saaty (1986, 1994). Variations in terminology occur throughout the AHP-application literature. For example, the quantity Saaty (1986) defines as a *priority vector* is frequently referred to as a “weight vector” or “weights”. We employ *priority vector* here.

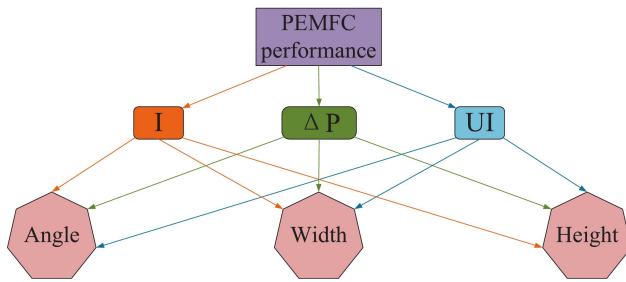
Specification of the evidentiary span.

Figure 2 reproduces the *hierarchy model* employed by Bu, et al (2026) for their fuel-cell design. Visual depiction — as a tree graph — of the decision-element hierarchy aids in understanding by human decision-makers. Idiosyncratic representative style notwithstanding, Figure 2 follows the ubiquitous form appearing in Saaty (1986, 1994), and throughout the MCDM literature. Structural elements of Figure 2 map to key DQ links from Figure 1.

- The rectangular box “PEMFC performance” at the top represents the *Goal or Focus*, in Saaty’s (1986) terms. A clear goal or focus results from thoughtful framing, in DQ terms.
- The curve-cornered boxes represent *Criteria*. Corresponding to DQ’s *values*, these define the span of measurable criteria believed to materially distinguish between alternatives. For this case, they represent device physical-operating parameters *current density I* , *pressure drop ΔP* , and *Oxygen-uniformity index UI* . These physical-operating parameters decisively influence the objective.
- The heptagonal shapes at the bottom depict decomposed criteria. Saaty (1986) describes “...intermediate set of higher criteria that can both be decomposed into these attributes and are themselves decompositions of the higher level criteria or subcriteria...”. In this case, each of three physical-device design specifications is believed to influence the operating parameters above.

Figure 2 omits the discrete alternatives from which the decision maker selects. Hierarchy-model depictions often include the alternatives when the set is small. Bu, et al (2026) however computationally simulate device performance based on 36 distinct combinations of geometric parameters — angle θ , width W , and height H — to construct distinct alternatives. The clutter arising from depicting each scenario in Figure 2 would detract from its clarity. Using AHP to analyze the risk of a large portfolio of distinct assets exemplifies another scenario in which the alternatives do not appear in the hierarchy.

Figure 2: Hierarchy model used for proton exchange membrane fuel cell (PEMFC) case study. (From Bu, et al (2026), Figure 5)



The criterion hierarchy in Figure 2 is also idiosyncratic for another reason: Each of the physical-specification sub-criteria influences each operating-parameter criteria. Typically, constituent subcriteria are distinct to their associated criterion. Their representations appear as “clean” tree graph. This excursion from common practice does not invalidate AHPs application, here. One might alternatively consider the Analytic Network Process (ANP) for scenarios like this (Saaty & Vargas (2013); Chen, et al (2019)).

Finally, Figure 2 includes an uncommon feature that clarifies interpretation: It shows edge directivity. Always directed tree graph, representation of AHP hierarchy models tend to omit directional indication. This reinforces our own graphical interpretation of AHP.

Comparative judgements about relative importance of criteria.

AHP imbues values and tradeoffs into decision-making through comparative judgements: “...a matrix to carry out pairwise comparisons of the relative importance of the elements [criteria]...” (emphasis added) (Saaty (1986)). Pairwise is the operative word, here. Practically, one assembles a list of unique, unordered element pairs for each hierarchy model. If the hierarchy model for a given objective (or criterion) spans n elements, then $\binom{n}{2}$ element pairs must be considered.

Figure 2 contains four hierarchic models, each based on three constituent elements. Consequently, $\binom{3}{2} = 3$ element pairs must be considered for each. Each element pair corresponds to a directed edge for the tournament graph by which the hierarchy model is represented.

Table 1 contains the comparative judgements assigned by Bu, et al (2026) for each hierarchy-model element pair depicted in Figure 2. We defer construction of corresponding matrices to the next subsection. Explicit representation here of the comparative judgements contributes clarity of the reasoning. Moreover, the directed-graph edge list in Table 1 provides a key of

departure for subsequent extension of the AHP methodology.

The “O-C” model corresponds to the highest level of the hierarchy model in Figure 2. It represents the contribution by each of the device operating parameters I , ΔP , and UI to overall performance of the PEMFC device. The constituent hierarchy depict the belief that geometric-design parameters angle θ , height H , and width W influence the operating parameters.

Now, (Saaty, 1986, Table 1) and (Saaty, 1994, Table 1) specifies a fundamental scale containing arbitrary, subjective heuristics to represent the intensity of importance between criterion pairs. (Bu, et al, 2026, Table 5) adopts this scale, as do many other AHP applications appearing in the literature. Saaty (1986), the most-rigorous description of the AHP methodology, contains no explicit justification as to why these specific values are necessary.

Since Saaty (1986) speaks to preferential ordering, alternative, more-rigorous enumerations of importance intensities can be applied. Application of the personal-probability indifference principle occurs as one alternative (Savage (1972); de Finetti (1974)). Hubbard & Seiersen (2023) — addressing the specific domain of cybersecurity — describe more-empirical approaches to measurement. Such techniques could similarly be adapted to AHP comparative judgements. Application of such alternatives to Saaty’s fundamental scale deflects a common critique of AHP’s logical rigor.

The construction of Table 1 anticipates its use in defining a directed graph in the sequel. The AHP literature — Saaty (1986, 1994, 2003), in particular — prescribes nothing about the ordering of element pairs in comparative judgement. Construction of Table 1 employed the following convention.

- Populate the “intensity of importance” column with the value greater than unity.
- Assign to the “reference element” column the most-preferred element, associated with the preferred intensity-of-importance value. Subsequently, element corresponds to the head of the directed graph edge connecting the two elements.
- Populate the “comparative element” column the less-preferred element. This element corresponds to the tail of the directed-graph edge.

This convention imbues two conveniences. First, the structure of the table endogenously contains the directive structure of the graph. This simplifies the algorithmic logic used to construct the resultant tournament graph. Second, it inclines us to think of importance intensities in terms as personal probabilities (Savage

Table 1: Comparative judgements for each hierarchy-model criterion pair. (After Bu, et al (2026), Tables 8, 9, 10, and 11)

hierarchy model	reference element	comparative element	intensity of importance
O-C	I	ΔP	7
O-C	I	UI	4
O-C	UI	ΔP	3
C- ΔP	H	W	7
C- ΔP	H	θ	5
C- ΔP	θ	H	3
C- UI	θ	H	5
C- UI	W	H	5
C- UI	W	θ	3
C- I	W	θ	5
C- I	W	H	3
C- I	H	θ	2

(1972); de Finetti (1974)) in lieu of the problematic “fundamental scale” from Saaty (1994).

Transformation of comparative judgments into priority vectors.

AHP priority vectors provide the key quantitative mechanism for evidentiary aggregation in MCDM applications. Mechanically, this is accomplished by constructing a reciprocal matrix from the comparative judgements for each hierarchy model. A reciprocal matrix is a Hermetian matrix with unit-valued diagonals and intensity-importance values assigned to off-diagonal elements. For the O-C model in Table 1, this appears as

$$A_{O-C} = \begin{bmatrix} 1 & a_{I,\Delta P} & a_{I,UI} \\ a_{\Delta P,I} & 1 & a_{\Delta P,UI} \\ a_{UI,I} & a_{UI,\Delta P} & 1 \end{bmatrix} \quad (1a)$$

$$= \begin{bmatrix} 1 & 7 & 4 \\ 1/7 & 1 & 1/3 \\ 1/4 & 3 & 1 \end{bmatrix}. \quad (1b)$$

Equation (1a) follows from Table 1 and

$$a_{j,i} = a_{i,j}^{-1},$$

by which we avoid logical contradiction in our reasoning. If

$$A_{O-C} X_{O-C} = X_{O-C} \Lambda_{O-C}, \quad (2)$$

represents the eigenvector/eigenvalue decomposition of reciprocal matrix A_{O-C} , then the resulting priority vector ω_{O-C} becomes the Perron (or principal) eigenvector (Saaty (2003)):

$$[\omega_{O-C}]_i = \frac{[X_{O-C}]_{i, \arg \max(\Lambda_{O-C})}}{\sum_i [X_{O-C}]_{i, \arg \max(\Lambda_{O-C})}}. \quad (3)$$

Table 2a contains the reciprocal matrix A_{O-C} and the priority vector ω_{O-C} resulting from the comparative judgements for the PEMFC-performance hierarchy model. The results coincide — within roundoff precision — with those reported by Bu, et al (2026).

Table 2a: Reciprocal matrix and priority vector resulting from A_{O-C} .

reference element	I	UI	ΔP	ω_i
I	1	4	7	0.705
UI	$1/4$	1	3	0.211
ΔP	$1/7$	$1/3$	1	0.084

Table 2b contains the same information for the constituent hierarchy models, $C-I$, $C-UI$, and $C-\Delta P$. The results do not coincide as closely with those from Bu, et al (2026), which are not reported with high degrees of numerical precision. The python *scipy* library (Virtanen, et al (2020)) produced the ω_i values appearing in Table 2b. Bu, et al (2026) do not report how their calculations were performed. Lacking these details, exact reproduction proves difficult.

Table 2b: Reciprocal matrices and priority vectors from $A_{C-\Delta P}$, A_{C-I} , A_{C-UI} .

hierarchy model	element	θ	W	H	ω_i
$C-I$	H	2	$1/3$	1	0.230
$C-I$	W	5	1	3	0.648
$C-I$	θ	1	$1/5$	$1/2$	0.1220
$C-UI$	H	$1/5$	$1/5$	1	0.0856
$C-UI$	W	3	1	5	0.618
$C-UI$	θ	1	$1/3$	5	0.230
$C-\Delta P$	H	5	7	1	0.740
$C-\Delta P$	W	$1/2$	1	$1/7$	0.0938
$C-\Delta P$	θ	1	2	$1/5$	0.1666

The mechanics illustrated by equations (1a, 2, and 3) are completely general. That ω_{O-C} is a projection onto the unit simplex Δ^3 where

$$\Delta^n = \left\{ x \in \mathbb{R}^n \mid \sum_i x_i = 1 \wedge x_i \geq 0 \forall i \right\}$$

is highly significant. All AHP calculation results are constrained to the interval $[0, 1]$. This includes both AHP-calculation results and standardization of inputs demonstrated in the next subsection.

Entropy of edge-weighted, directed acyclic graph.

Data availability.

This GitHub repository <https://github.com/hamlett-neil-ur/ahp-graph-entropy> contains the data and the python and Neo4j logic on which this paper is based.

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